Apache Parquet

1- Reading and Writing using Pandas

EXPLORER

UNTITLED (WORKSPACE)

- python testproject
  - .vscode
  - data.parquet
  - test.py

Welcome | test.py

python testproject > test.py > ...

```python

import pandas as pd
#1
#writing a parquet file
data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35],
    "City": ["New York", "Los Angeles", "Chicago"]
}
df= pd.DataFrame(data)
df.to_parquet("data.parquet",engine="pyarrow", index=False)
print("Parquet file written successfuly!")

#reading a parquet file
df_read_file= pd.read_parquet("data.parquet",engine="pyarrow")
print(f"Data from parquet file \n{df_read_file}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\USER\Desktop\vsCode projects\python testproject>
PS C:\Users\USER\Desktop\vsCode projects\python testproject> & C:\Users\USER\AppDa
Desktop/vsCode projects/python testproject/test.py"
Parquet file written successfuly!
Data from parquet file
      Name  Age         City
0    Alice   25     New York
1      Bob   30  Los Angeles
2  Charlie   35      Chicago
PS C:\Users\USER\Desktop\vsCode projects\python testproject>
```

2- Reading and Writing using PyArrow

EXPLORER

UNTITLED (WORKSPACE)

- python testproject
  - .vscode
  - data.parquet
  - test.py

Welcome | test.py | data.parquet

python testproject > test.py > ...

```python
import pandas as pd
import pyarrow as pa
import pyarrow.parquet as pq

#2 using PyArrow

#writing parquet file
df = pd.DataFrame({
    "Name": ["Jacob", "Lauren", "Oliver"],
    "Age": [25, 30, 35],
    "City": ["New York", "Los Angeles", "Chicago"]
})
table = pa.Table.from_pandas(df)
pq.write_table(table, "data.parquet")
print("Parquet file written successfully!")

#reading parquet file
table= pq.read_table("data.parquet")
pd_read= table.to_pandas()
print(f"Data from Parquet file: \n{pd_read}")
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\USER\Desktop\vsCode projects\python testproject> & C:\
Desktop/vsCode projects/python testproject/test.py"
Parquet file written successfully!
Data from Parquet file:
     Name  Age         City
0   Jacob   25     New York
1  Lauren   30  Los Angeles
2  Oliver   35      Chicago
```

3- Using PySpark

```python
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("SparkExample").getOrCreate()

data = [
    ("Jacob", 30, "New York"),
    ("Lauren", 35, "Los Angeles"),
    ("Billy", 25, "Chicago")
]

schema = ["Name", "Age", "City"]

df = spark.createDataFrame(data, schema)
df.show()
```

```
+------+---+-----------+
|  Name|Age|       City|
+------+---+-----------+
| Jacob| 30|   New York|
|Lauren| 35|Los Angeles|
| Billy| 25|    Chicago|
+------+---+-----------+
```

```python
df.write.parquet("data.parquet")
parquet_df = spark.read.parquet("data.parquet")
parquet_df.show()
```

```
+------+---+-----------+
|  Name|Age|       City|
+------+---+-----------+
|Lauren| 35|Los Angeles|
| Billy| 25|    Chicago|
| Jacob| 30|   New York|
+------+---+-----------+
```

4- Append Data to an existing Parquet file + Read Specific Columns

EXPLORER

UNTITLED (WORKSPACE)

- python testproject
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Welcome | test.py | data.parquet

python testproject > test.py > ...

```python
import pyarrow.parquet as pq
import pyarrow as pa
import pandas as pd
# Loading existing Parquet file
existing_table = pq.read_table("data.parquet")

new_data = pd.DataFrame({
    "Name": ["David", "Emma"],
    "Age": [40, 28],
    "City": ["San Francisco", "Seattle"]
})

new_table = pa.Table.from_pandas(new_data)

# concatenating both tables
merged_table = pa.concat_tables([existing_table, new_table])
# writing back to Parquet file
pq.write_table(merged_table, "data.parquet")

#Reading specific columns from a Parquet file
df = pd.read_parquet("data.parquet", columns=["Name", "Age"])
print(df)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\USER\Desktop\vsCode projects\python testproject> & C:\Users\USER
Desktop/vsCode projects/python testproject/test.py"
     Name  Age
0   Jacob   25
1  Lauren   30
2  Oliver   35
3   David   40
4    Emma   28
PS C:\Users\USER\Desktop\vsCode projects\python testproject>
```

5- Filtering on Age

Welcome | test.py | data.parquet

python testproject > test.py > ...

```python
import pyarrow.parquet as pq

# Reading only rows where Age > 30
table = pq.read_table("data.parquet", filters=[("Age", ">", 30)])

df = table.to_pandas()
print(df)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\USER\Desktop\vsCode projects\python testproject> & C:\Users\USER\AppDa
Desktop/vsCode projects/python testproject/test.py"
     Name  Age           City
0  Oliver   35        Chicago
1   David   40  San Francisco
```

6- Merge Multiple Parquet Files

Welcome | test.py

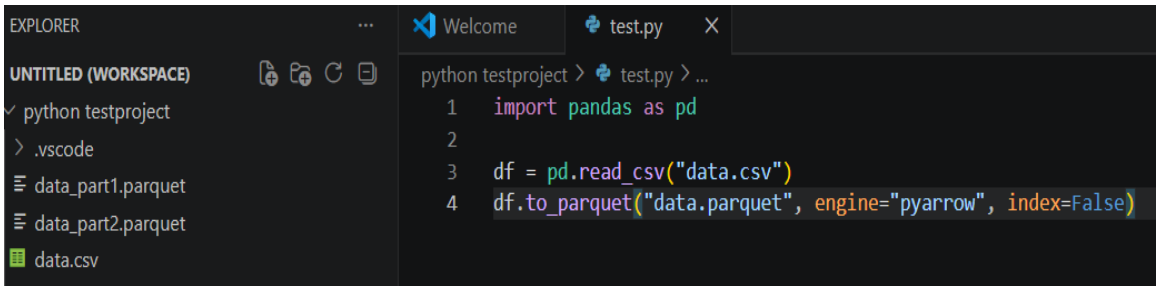
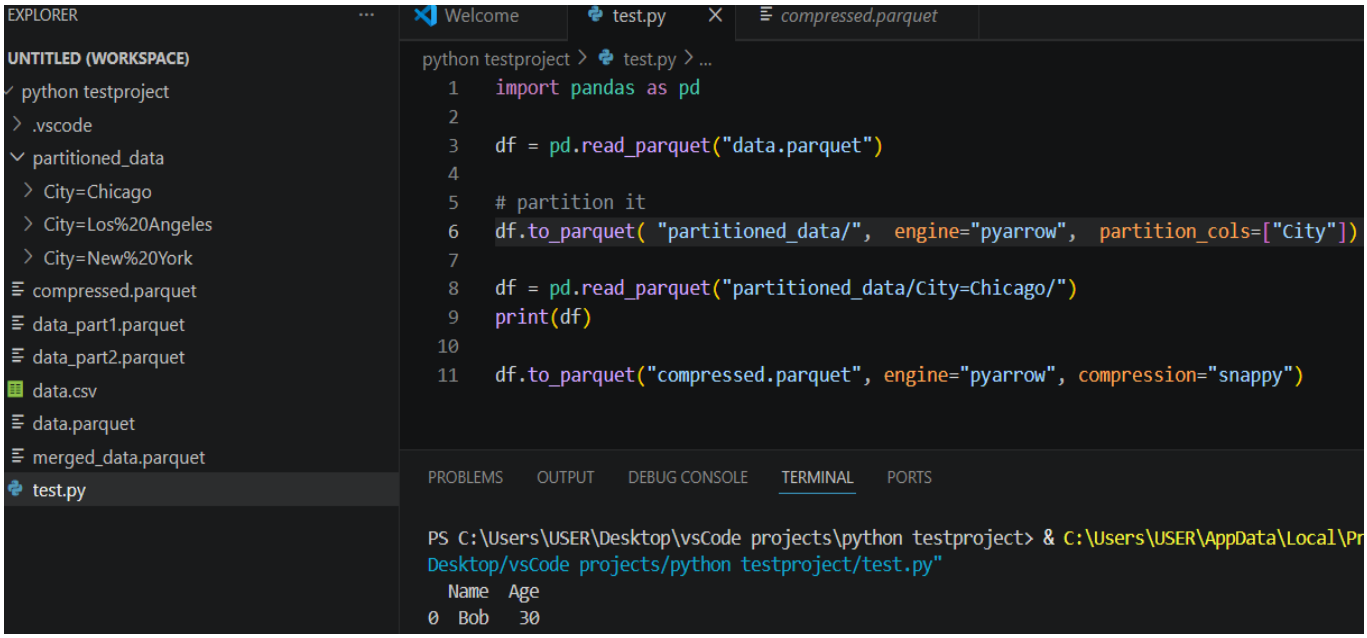
python testproject > test.py > ...

```python
#samples
df1 = pd.DataFrame({
    "Name": ["Alice", "Bob"],
    "Age": [25, 30],
    "City": ["New York", "Chicago"]
})

df2 = pd.DataFrame({
    "Name": ["Charlie", "David"],
    "Age": [35, 40],
    "City": ["New York", "Los Angeles"]
})
df1.to_parquet("data_part1.parquet", engine="pyarrow",index=False)
df2.to_parquet("data_part2.parquet", engine="pyarrow",index=False)

file_list = ["data_part1.parquet", "data_part2.parquet"]
tables=[ pq.read_table(f) for f in file_list]
merged_table = pa.concat_tables(tables)

pq.write_table(merged_table, "merged_data.parquet")

read_merged = pq.read_table("merged_data.parquet")
df_read_merged = read_merged.to_pandas()

print("Data from merged Parquet files:")
print(df_read_merged)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\USER\Desktop\vsCode projects\python testproject> & C:\Users\USER\AppData\Lo
Desktop/vsCode projects/python testproject/test.py"
Data from merged Parquet files:
      Name  Age         City
0    Alice   25     New York
1      Bob   30      Chicago
2  Charlie   35     New York
3    David   40  Los Angeles
```

7- Convert CSV to Parquet



8- Partition Parquet Files + Compression